Santiago Giraldo Godoy

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EDUCATION

Northeastern University (NU)
Master of Science in Artificial Intelligence, Concentration in Machine Learning **May 2028**

Worcester Polytechnic Institute (WPI)
Bachelor of Science in Computer Science, Minor in Robotics Engineering **May 2026**
Graduated with High Distinction | GPA: 3.83/4.0 | Upsilon Pi Epsilon (UPE) Honor Society
Data Science Outstanding MQP Award (2026) – Awarded for AI for Impact: Inferring Suicidal Ideation via Smartphones

SKILLS

Programming Languages: Java, TypeScript, C++, C, JavaScript, CSS, HTML, PL/SQL, PostgreSQL, Kotlin, Python, Swift
AI / Machine Learning: PyTorch, Deep Learning, CNNs, Machine Learning Model Training, Prompt Engineering, RAG tools, AI Agents
Software Frameworks: React.js, Shadcn UI, Express, Tailwind, Bootstrap, Node.js, JWT, MERN Stack
Languages: Spanish: Native, English: Bilingual proficiency

EXPERIENCE

Software Engineer Intern, Oracle, Redwood Shores **May 2025 – Aug 2025**
• Improved AI agent response quality through **prompt engineering and optimization of Retrieval-Augmented Generation (RAG)** workflows, increasing **human-like response rates by 50%**.
• Migrated and optimized data unification from multiple sources into a single, standardized database through SQL and Java endpoints allowing more than a data transfer improvement **from 6 sec/entry to 0.2 sec/entry, shortening migration by 97%.**
• Created multiple Java endpoints to handle data filtration, error handling, and data storage and retrieval **causing 25% more accuracy**.
• Improved AI chatbot responses by unifying databases, improving prompt handling and allowing user feedback to be stored for review, resulting in a beta version of a troubleshooting agent meant to **reduce expert workload and resource consumption by 60%.**

AI Project worker, WPI, Worcester **Jan 2023 – Dec 2024**
• Taught an AI program with over **300 images per week**, enhancing the machine vision capabilities of the AI.
• Trained AI software to safeguard 5th grade students' confidential information, ensuring data privacy and security for over 1000 students.

Chief Operating Officer, Universal Education Initiative (UEI), Worcester **Sep 2023 – Sep 2024**
• Managed day-to-day activities of the organization, ensuring correct synchronization by 5 teams, which improved operational efficiency
• Developed timelines, established roles and responsibilities, and set company objectives, leading to improved company performance and goal alignment
• Tracked different company teams' performance and created biweekly reports for the CEO, which facilitated informed decision-making and strategic planning

ACADEMIC PROJECTS

AI for Impact: Inferring Suicidal Ideation via Smartphones, WPI **Aug 2025 – May 2026**
Recipient of WPI's 2026 Data Science Outstanding MQP Award
• Refactored and re-architected a Kotlin Multiplatform (KMP) mental health monitoring app to support up to **80% shared Android/iOS codebases,** ensuring maintainability and cross-platform parity.
• Implemented passive smartphone data collection (motion, app usage, Bluetooth, survey data) for multi-modal behavioral analysis during an active medical research trial.
• Integrated and tested Bluetooth and background data collection workflows to ensure consistent sensor data acquisition across platforms.
• Conducted exploratory data analysis and feature engineering to correlate digital behavior patterns with self-reported suicidal ideation.

Spreadsheet application, WPI **Jan 2026 – Mar 2026**
• Designed and implemented a full-stack spreadsheet application, following the **MVC architectural pattern** to separate logic.
• Developed an **extensible** evaluation engine using **design patterns**, **dynamic formula computation** and **automatic propagation cells**.
• Wrote comprehensive **JUnit** tests to validate expression evaluation, parsing logic, and edge cases, ensuring reliability of the system.

Traffic Sign Recognition Models, Machine Learning, WPI **Oct 2024 – Dec 2024**
• Developed a machine learning model for traffic sign classification, achieving **99.50% accuracy** using a custom CNN.
• Fine-tuned pre-trained models (ResNet50, VGG16) and optimized hyperparameters for improved performance from **70% to +95%.**
• Implemented data preprocessing and augmentation techniques to enhance model generalization.

Medical App Prototype for Brigham and Women's Hospital, Software Engineering, WPI **Mar 2024 – May 2024**
• Worked on a **10-person team to apply Agile** methodologies and software design patterns in the creation of a web application prototype.
• Developed the web application using React.js, Typescript, Shadcn UI, Express, Prisma ORM, and PostgreSQL that included hospital pathfinding, a map editor, a functional EMR input and more.
• Designed backend of the application through the creation of ERD and writing the Prisma schema components creating **90% of schema.**
• Served as the technical leader of the backend subgroup handling the tRPC routes to connect the frontend and backend of the app.